

Homework associated to L04 stochastic simulation of a model

In order to understand which are the key elements of the stochastic simulation, perform the following exercises exploiting the simulator `pySSA.py` that you can find in the `ClassMaterial` folder.

Keep in mind that the syntax to call the program is:

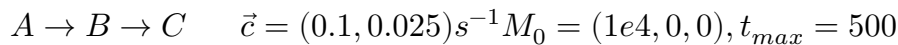
```
python pySSA.py modelFileName tEnd
```

where: * `modelFileName` is the model definition file written according to the syntax described in L03 homework, * `tEnd` is the simulation halting time.

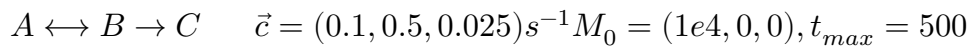
Exercises:

Run the simulations for the following models, with the given initial amounts and halting time:

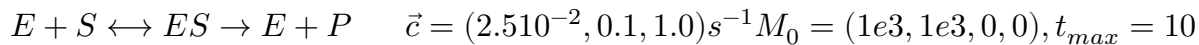
- **Cascade**



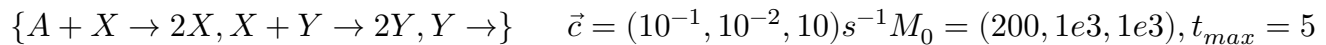
- **Cascade with equilibrium**



- **Michaelis Menten**



- **Lotka-Volterra**



Exercise: simulations fo the Prokaryotic auto-regulatory gene network

from [Wang et al. BMC Systems Biology 2010, 4:99](#)

Run the simulations for the following model, with the given initial amounts:

The model we tested is a prokaryotic auto-regulatory gene network in which dimmers of a protein repress its own gene transcription by binding to a regulatory region upstream of the gene. The system, involving both transcription and translation, can serve as a simple, yet illustrative, example of gene regulation [31,39].

The reactions in the network are given below:

```
R1 : DNA + P2 → DNA.P2,
R2 : DNA.P2 → DNA + P2,
R3 : DNA → DNA + mRNA,
R4 : mRNA → ,
R5 : 2P → P2,
R6 : P2 → 2P,
R7 : mRNA → mRNA + P,
R8 : P → .
```

Here DNA, P, P2 and mRNA represent promoter sequences, proteins, protein dimmers and messenger RNA respectively. In this model, mRNAs and proteins are synthesized by transcription and translation processes (R3 and R7), and destroyed by degradation (R4 and R8). The proteins can form a dimmer P2 (R5 and R6), which binds and unbinds to DNA (R1 and R2). When a protein dimmer binds to the promoter, it represses mRNA production. Overall, the network implements a self-regulatory mechanism to control the synthesis of the protein product, suppressing the transcription when the protein product is abundant. Note that $\text{DNA} + \text{DNA.P2}$ is a conserved quantity in the system.

... within a time window of [0 50) with $(k_1, \dots, k_8) = (0.1, 0.7, 0.35, 0.3, 0.1, 0.9, 0.2, 0.1)$.

And $\text{DNA} = 2$

Exercise: simulations for the repressilator

from Elowitz, M., Leibler, S. A synthetic oscillatory network of transcriptional regulators. *Nature* 403, 335–338 (2000)

Run the simulations for the following model:

... We used three transcriptional repressor systems that are not part of any natural biological clock to build an oscillating network, termed the repressilator, in *Escherichia coli*. The network periodically induces the synthesis of green fluorescent protein as a readout of its state in individual cells.

Network construction

The three repressors (*lacI*, *tetR* and *cl*), the λ promoter P_R and the unstable variants of *gfp* were all cloned by the polymerase chain reaction (PCR) and verified by sequencing. In the case of *lacI*, the GTG start codon was changed to ATG with the 5' primer. 39' PCR primers were used to add the coding sequence for the 11-amino-acid *ssrA* tag to the three repressors. Tagging of full-length λ repressor, as well as *lacI* and *tetR*, resulted in functional proteins that required increased induction to achieve full repression, consistent with decreased stability. Promoter sequences P_{Llac01} and P_{Ltet01} were obtained from *pZE21-MCS1* and *pZE12-luc*. The repressilator plasmid was constructed in three successive cloning steps from intermediate plasmids containing the three transcriptional units. The reporter plasmid was constructed by cloning the *GFPaav* variant, which has a half-life of around 90 min, into *pZE21-MCS1* (ref. 6). The finite rate at which *GFP* is oxidized to its fluorescent form and its relatively long half-life limit our ability to observe fast dynamic phenomena. The presence of additional *tet* operators on the reporter plasmid constitutes a perturbation to the single-plasmid repressilator network by titrating out repressors.

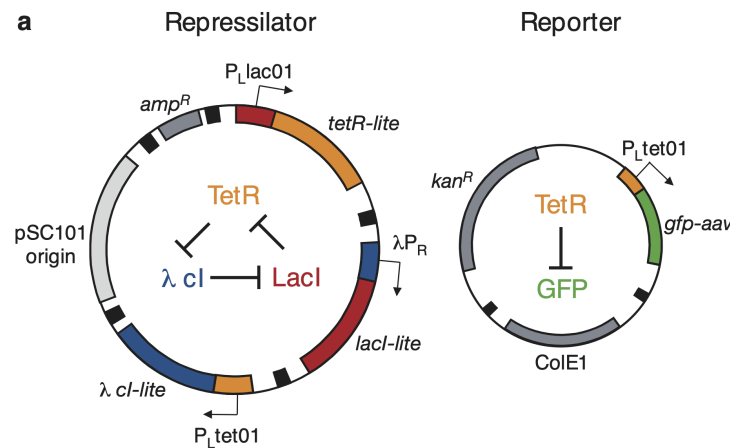


Figure 1: The model